1) Before understanding this concept, read **Bagging** first.
2) Random forest is Bagging with default “estimator = Decision Tree” but with a new feature which we’ll know soon.

3) The relationship between bias and variance is inverse. For models with Low Bias and High Variance, both Bagging and Random forest helps to reduce the variance but Random Forest reduces more than Bagging. Which is explained in “**difference_between_bagging_and_randomForest.ipynb**” with visualization proof.

4) How Random forest reduces variance?

   Now a model returns output of Low Bias and High Variance when there are a lot of noises in the dataset. When we use one model, Decision Tree, it takes all the noises, learns all of them, and returns very good results on the training dataset but not on the testing dataset. But when we use Random Forest, collections of Decision Trees, we give each one of them a sample and along with the sample the noises also get distributed and often one model gets most of the noises for what the other decision tree models with lesser noise returns output of low variance.

   How bias remains low for both one decision tree and Random Forest? Because decision trees perform well on training dataset, so the bias is always low but due to the noises they perform less on testing dataset i.e. high variance and when we distribute those noises in Random Forest, the variance becomes low in the overall output of Random Forest.